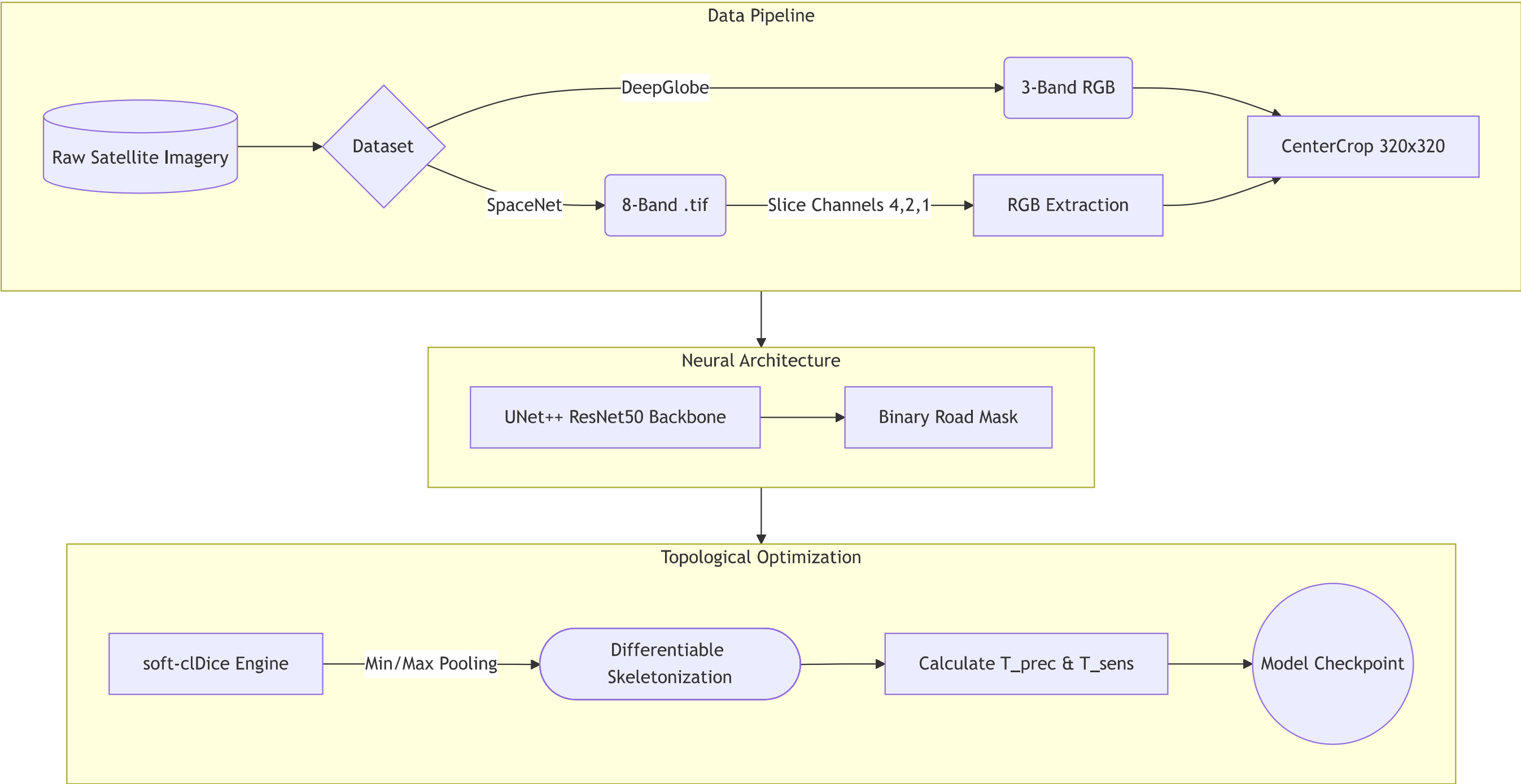


Topology-Preserving Road Extraction (ISRO Hackathon)

This repository contains an end-to-end deep learning pipeline designed to extract complex road networks from high-resolution satellite imagery. It is built to natively support **SpaceNet 5** (Mumbai, Moscow) and **DeepGlobe** datasets.

Project Flowchart



The Core Engine: **soft-clDice** Loss

Extracting tubular structures like roads or blood vessels is a uniquely difficult computer vision problem. Standard loss functions (like Cross-Entropy or standard Dice) are calculated pixel-by-pixel. If a model misses a single pixel that connects a 5-mile highway, standard Dice barely penalizes it because it's "just one pixel." However, in reality, the road network's **topology is fundamentally broken**.

To solve this, we implemented **soft-clDice** (Centerline Dice), introduced in the CVPR 2021 paper: "*clDice - A Novel Topology-Preserving Loss Function for Tubular Structure Segmentation*".

1. Soft-Skeletonization (Algorithm 1)

To penalize broken roads, the model must understand the "centerline" or "skeleton" of the road network. However, traditional morphological skeletonization algorithms (like those in Scikit-Image) are non-differentiable—meaning a neural network cannot learn or backpropagate through them.

Our pipeline implements the paper's differentiable proxy for skeletonization using **iterative min/max pooling**: * **Min Pooling (Erosion)**: Shrinks the road. * **Max Pooling (Dilation)**: Expands it back out. * By subtracting the morphologically opened image from the original, we extract the skeletal structure purely using PyTorch tensors, allowing gradients to flow freely during training!

2. Topological Metrics (Algorithm 2)

Once we have the differentiable skeletons for both the Ground Truth (S_L) and the Prediction (S_P), we calculate: * **Topology Precision (T_{prec})**: The percentage of the predicted skeleton that accurately falls inside the true road mask. * **Topology Sensitivity (T_{sens})**: The percentage of the true skeleton that is successfully covered by the predicted road mask.

The final **soft-clDice** score is the harmonic mean of T_{prec} and T_{sens} . The total loss function combines standard soft-Dice (for volumetric accuracy) and soft-clDice (for topological connectivity) using a weighting factor of $\alpha = 0.45$.

Features Implemented

- 1. **Universal RGB Compatibility**: SpaceNet 5 uses 8-band multispectral imagery, which historically locked models into requiring specialized data. We implemented an on-the-fly tensor slicing algorithm that strips the infrared data, forcing the model to become an absolute master at standard RGB road extraction. This makes the model universally compatible with Google Maps, Drones, and the DeepGlobe dataset.
- 2. **Transfer Learning Pipeline**: The architecture is designed to train on one city (e.g., Mumbai), automatically save its weights, and dynamically load those weights to continue fine-tuning on a completely different geography (e.g., Moscow or DeepGlobe).
- 3. **Continuous Checkpointing**: The pipeline writes to the **.pth** file continuously at the end of every epoch. If a massive 15-hour training run crashes at hour 14, the weights from the previous epoch are safely preserved.
- 4. **Hardware Fallback**: Dynamically detects execution environments, scaling down to CPU-safe parameters for development on Intel Iris Xe laptops, while remaining instantly ready to utilize the Tensor Cores of an RTX 5060 when deployed.